

Chapter 13:

Data and Database Administration

Traditional Administration Definitions

- ***Data Administration*** A high-level function that is responsible for the overall management of data resources in an organization, including maintaining corporate-wide definitions and standards
- ***Database Administration*** A technical function that is responsible for physical database design and for dealing with technical issues such as security enforcement, database performance, and backup and recovery

Traditional Data Administration Functions

- Data policies, procedures, standards
- Planning
- Data conflict (ownership) resolution
- Managing the information repository
- Internal marketing of DA concepts

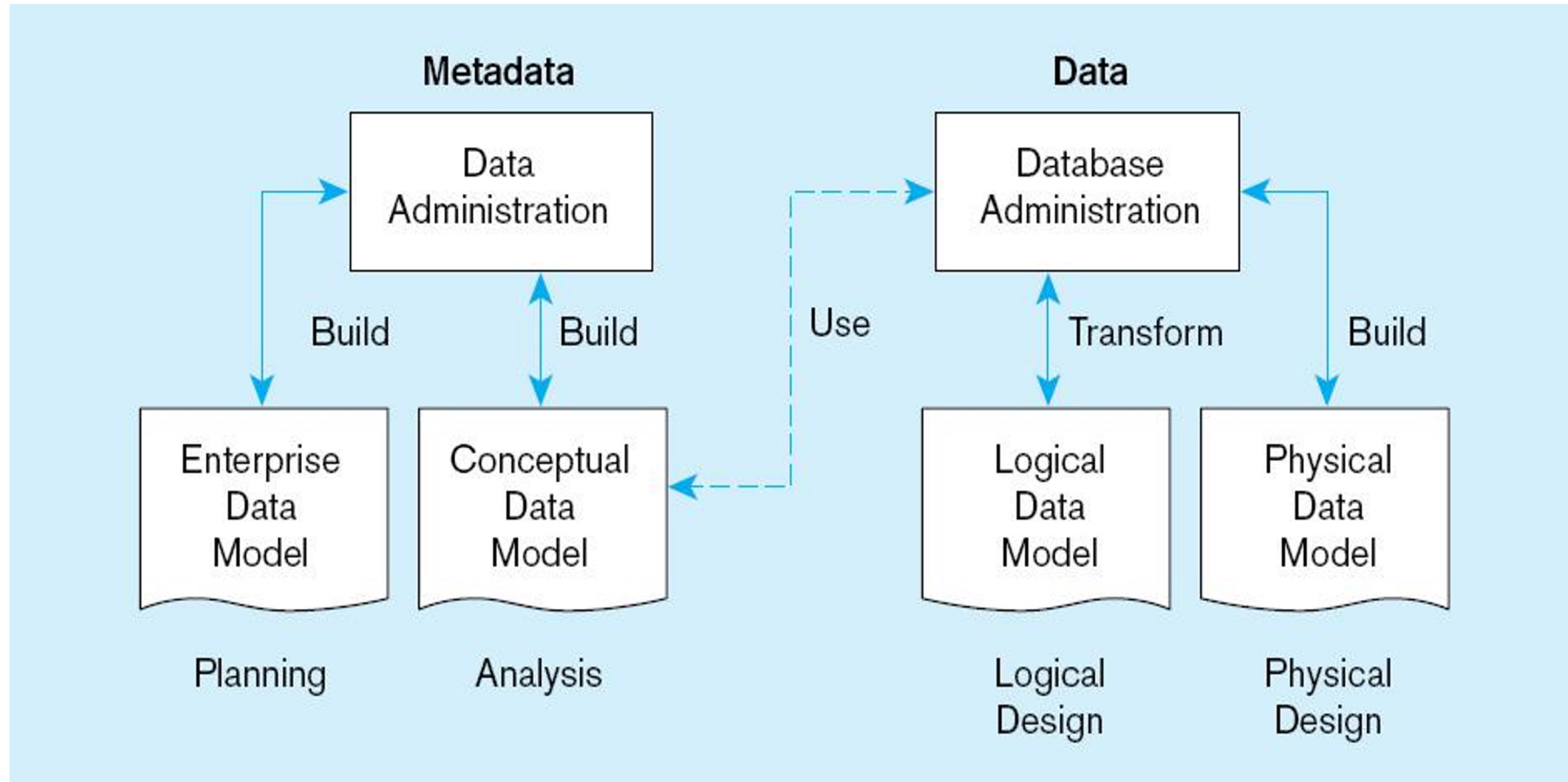
Traditional Database Administration Functions

- Selection of DBMS and software tools
- Installing/upgrading DBMS
- Tuning database performance
- Improving query processing performance
- Managing data security, privacy, and integrity
- Data backup and recovery

Data Warehouse Administration

- New role, coming with the growth in data warehouses
- Similar to DA/DBA roles
- Emphasis on integration and coordination of metadata/data across many data sources
- Specific roles:
 - Support DSS applications
 - Manage data warehouse growth
 - Establish service level agreements regarding data warehouses and data marts

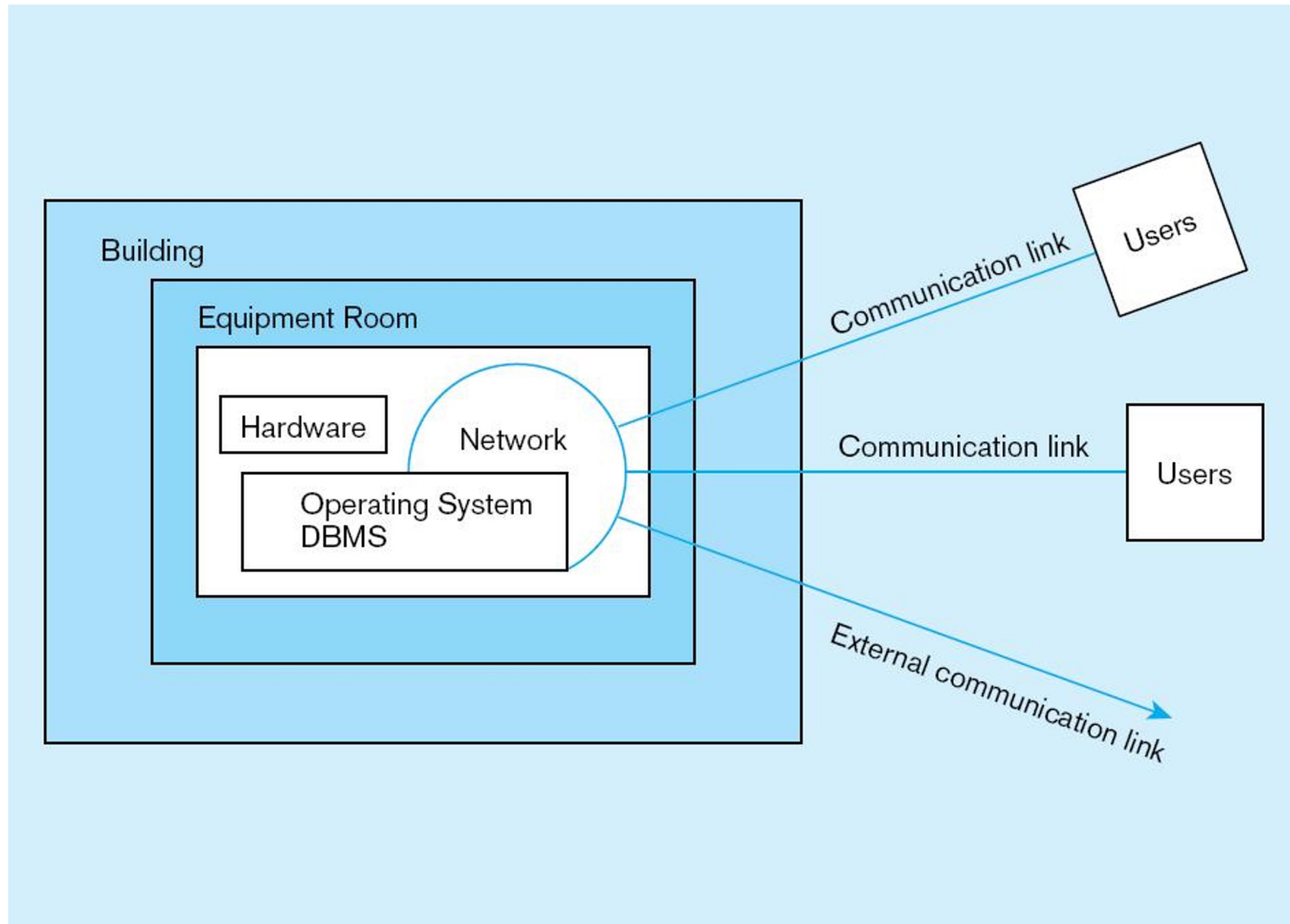
Figure 13-2 Data modeling responsibilities



Database Security

- **Database Security:** Protection of the data against accidental or intentional loss, destruction, or misuse
- Increased difficulty due to Internet access and client/server technologies

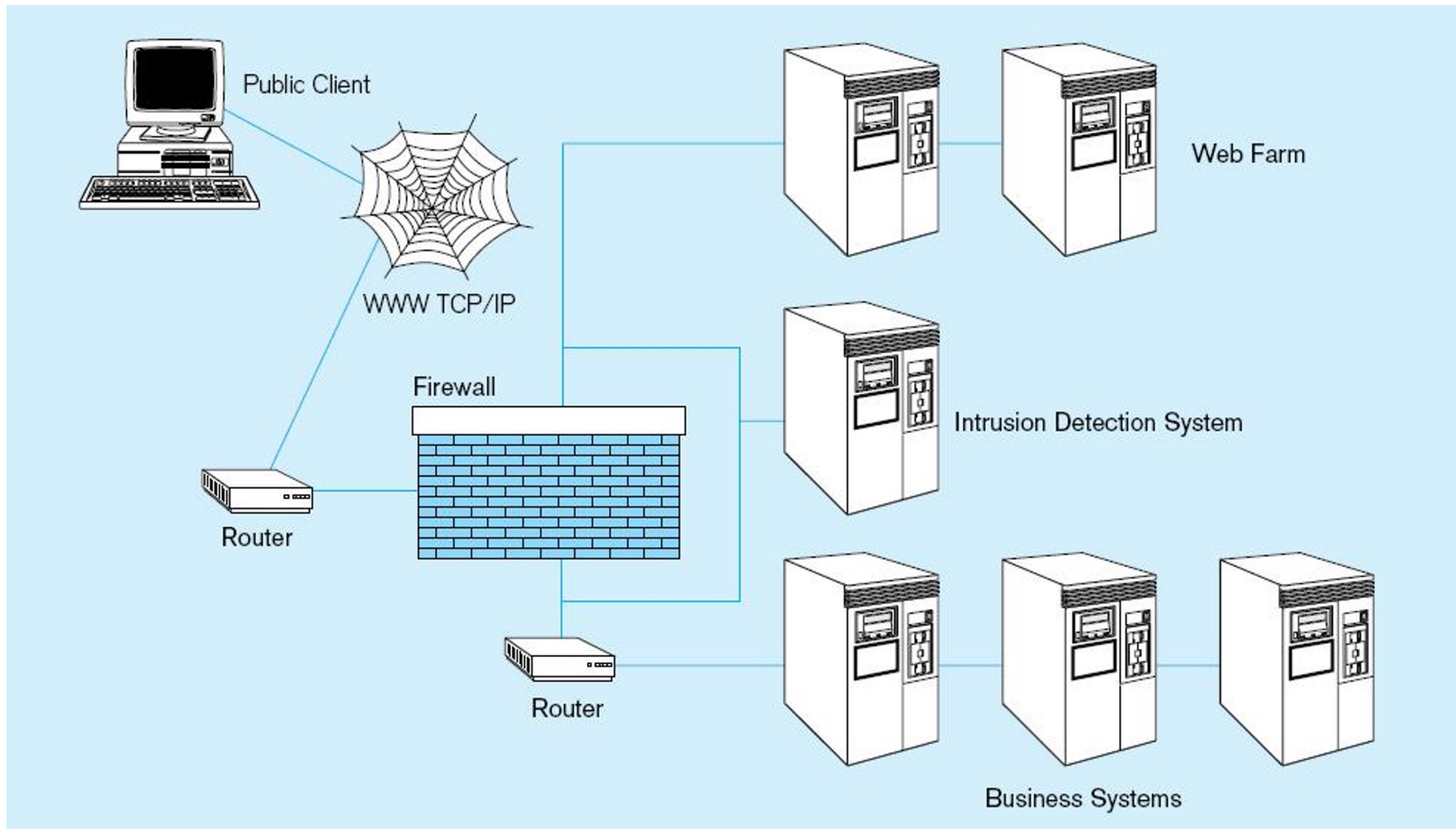
Figure 13-3 Possible locations of data security threats



Threats to Data Security

- Accidental losses attributable to:
 - Human error
 - Software failure
 - Hardware failure
- Theft and fraud
- Improper data access:
 - Loss of privacy (personal data)
 - Loss of confidentiality (corporate data)
- Loss of data integrity
- Loss of availability (through, e.g. sabotage)

Figure 13-4 Establishing Internet Security



Security Policies and Procedures

- Personnel controls
 - Hiring practices, employee monitoring, security training
- Physical access controls
 - Equipment locking, check-out procedures, screen placement
- Maintenance controls
 - Maintenance agreements, access to source code, quality and availability standards
- Data privacy controls
 - Adherence to privacy legislation, access rules

Database Recovery

Mechanism for restoring a database quickly and accurately after loss or damage

Recovery facilities